AN ISOLATED POINT OF A FINITE TYPE FIBRE BECOMES FINITE OVER AN ELEMENTARY ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11923-11953 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 11955-11963, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The characterization of quasi-finiteness at a point as an isolated fibre point under finite type (Morphisms, cited in the statement footnote) and the affine algebra case (Algebra, etale-makes-quasi-finite-finite-one-prime); cited inputs with the localization reduction written in the retained span.

Source reference. Original lemma 00UJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11963–11963. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11909–11916. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11963–11963. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$. Set $s = f(x)$. Assume that*

(1) *f is locally of finite type, and*
(2) *$x \in X_s$ is isolated*[1].

Then there exist

(a) *an elementary étale neighbourhood $(U, u) \to (S, s)$,*
(b) *an open subscheme $V \subset X_U$ (see 02LJ)*

such that

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

[1] In the presence of (1) this means that f is quasi-finite at x, see Morphisms, Lemma 01TH.

(i) *$V \to U$ is a finite morphism,*
(ii) *there is a unique point v of V mapping to u in U, and*
(iii) *the point v maps to x under the morphism $X_U \to X$, inducing $\kappa(x) = \kappa(v)$.*

Moreover, for any elementary étale neighbourhood $(U', u') \to (U, u)$ setting $V' = U' \times_U V \subset X_{U'}$ the triple (U', u', V') satisfies the properties (i), (ii), and (iii) as well.

Proof. Let $Y \subset X$, $W \subset S$ be affine opens such that $f(Y) \subset W$ and such that $x \in Y$. Note that x is also an isolated point of the fibre of the morphism $f|_Y : Y \to W$. If we can prove the theorem for $f|_Y : Y \to W$, then the theorem follows for f. Hence we reduce to the case where f is a morphism of affine schemes. This case is Algebra, Lemma 00UJ. □